

LAB EXPERIMENT – 4

Crypt-Arithmetic Problem using Python — CO2

Aim

To write a Python program to solve a **Crypt-Arithmetic problem**, where each letter represents a unique digit.

Problem

Solve:

SEND

+ MORE

MONEY

Procedure

1. Represent each alphabetic character by a unique digit from 0–9.
2. Ensure that the first letter of each word is not zero.
3. Generate possible digit assignments using permutations.
4. Substitute the digits into the given equation.
5. Check whether the arithmetic equation is satisfied.
6. Display the valid solution.

Source Code

main.py

Visualize

Run ▶

Output

```
1 from itertools import permutations
2
3 word1 = "SEND"
4 word2 = "MORE"
5 result = "MONEY"
6
7 letters = set(word1 + word2 + result)
8 letters = list(letters)
9
10 for digits in permutations(range(10), len(letters)):
11
12     mapping = dict(zip(letters, digits))
13
14     # Leading letters cannot be zero
15     if mapping["S"] == 0 or mapping["M"] == 0:
16         continue
17
18     send = (
19         mapping["S"] * 1000 +
20         mapping["E"] * 100 +
21         mapping["N"] * 10 +
22         mapping["D"]
23     )
24
25     more = (
26         mapping["M"] * 1000 +
27         mapping["O"] * 100 +
28         mapping["R"] * 10 +
29         mapping["E"]
30     )
31
32     money = (
33         mapping["M"] * 10000 +
34         mapping["O"] * 1000 +
35         mapping["N"] * 100 +
36         mapping["E"] * 10 +
37         mapping["Y"]
38     )
39
40     if send + more == money:
41         print("Solution found:")
42         print("SEND =", send)
```

Solution found:

SEND = 9567

MORE = 1085

MONEY = 10652

Letter-Digit Mapping:

D = 7

E = 5

M = 1

N = 6

O = 0

R = 8

S = 9

Y = 2

main.py

Visualize

Run ▶

```
8 letters = list(letters)
9
10 for digits in permutations(range(10), len(letters)):
11
12     mapping = dict(zip(letters, digits))
13
14     # Leading letters cannot be zero
15     if mapping["S"] == 0 or mapping["M"] == 0:
16         continue
17
18     send = (
19         mapping["S"] * 1000 +
20         mapping["E"] * 100 +
21         mapping["N"] * 10 +
22         mapping["D"]
23     )
24
25     more = (
26         mapping["M"] * 1000 +
27         mapping["O"] * 100 +
28         mapping["R"] * 10 +
29         mapping["E"]
30     )
31
32     money = (
33         mapping["M"] * 10000 +
34         mapping["O"] * 1000 +
35         mapping["N"] * 100 +
36         mapping["E"] * 10 +
37         mapping["Y"]
38     )
39
40     if send + more == money:
41         print("Solution found:")
42         print("SEND =", send)
43         print("MORE =", more)
44         print("MONEY =", money)
45         print("\nLetter-Digit Mapping:")
46
47         for letter in sorted(mapping):
48             print(letter, "=", mapping[letter])
49
50     break
```

Output

Solution found:
SEND = 9567
MORE = 1085
MONEY = 10652

Letter-Digit Mapping:
D = 7
E = 5
M = 1
N = 6
O = 0
R = 8
S = 9
Y = 2

Result

Thus, the **Crypt-Arithmetic** problem was successfully solved using Python.